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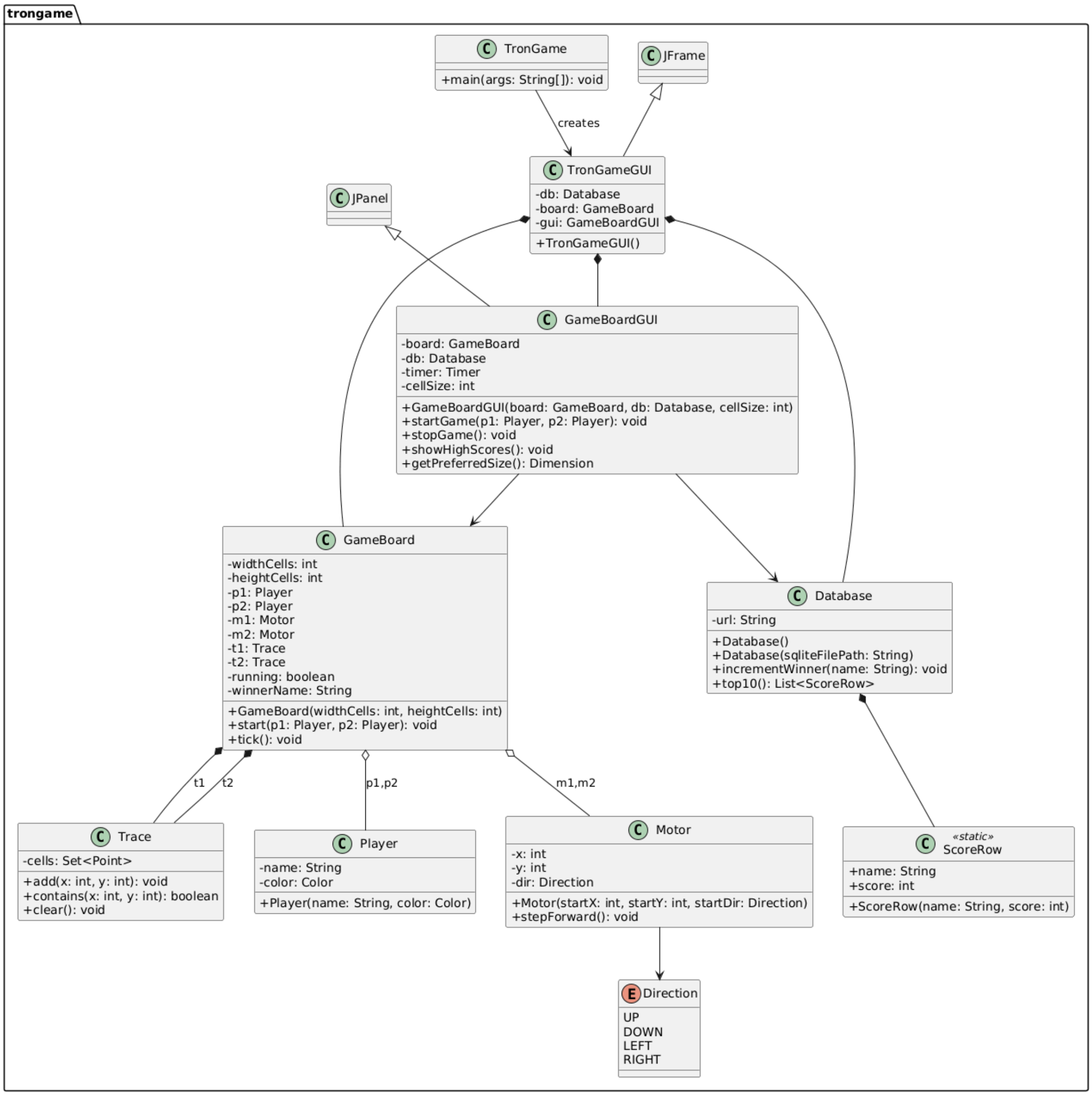
Task description

Tron Create a game, with we can play the light-motorcycle battle (known from the Tron movie) in a top view. Two players play against each other with two motors, where each motor leaves a light trace behind of itself on the display. The motor goes in each seconds toward the direction, that the player has set recently. The first player can use the WASD keyboard buttons, while the second one can use the cursor buttons for steering. A player loses if its motor goes to the boundary of the game level, or it goes to the light trace of the other player. Ask the name of the players before the game starts, and let them choose the color their light traces. Increase the counter of the winner by one in the database at the end of the game. If the player does not exit in the database yet, then insert a record for him. Create a menu item, which displays a highscore table of the players for the 10 best scores. Also, create a menu item which restarts the game.



The most interesting algorithms:

1. onTick():this Is the main logic of the game which runs in ticks, which happen every 80ms. On each tick the game advances motors 1 cell forward to their directions. Then checks for collision or leaving the bounds of the board. If one player lost, other player becomes winner and his score increments by 1 on the database. If both players lose on the same tick no one wins, it is a draw. If no one lost or won then it extends the trails of the motors.
2. setDir(): the second logic does not let motors instantly change their directions by 180*. If it is going up and the DOWN key is pressed it is not going to change it's direction.

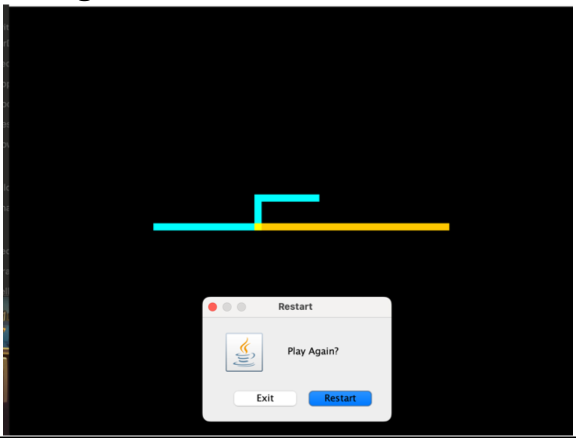
Events and Handlers:

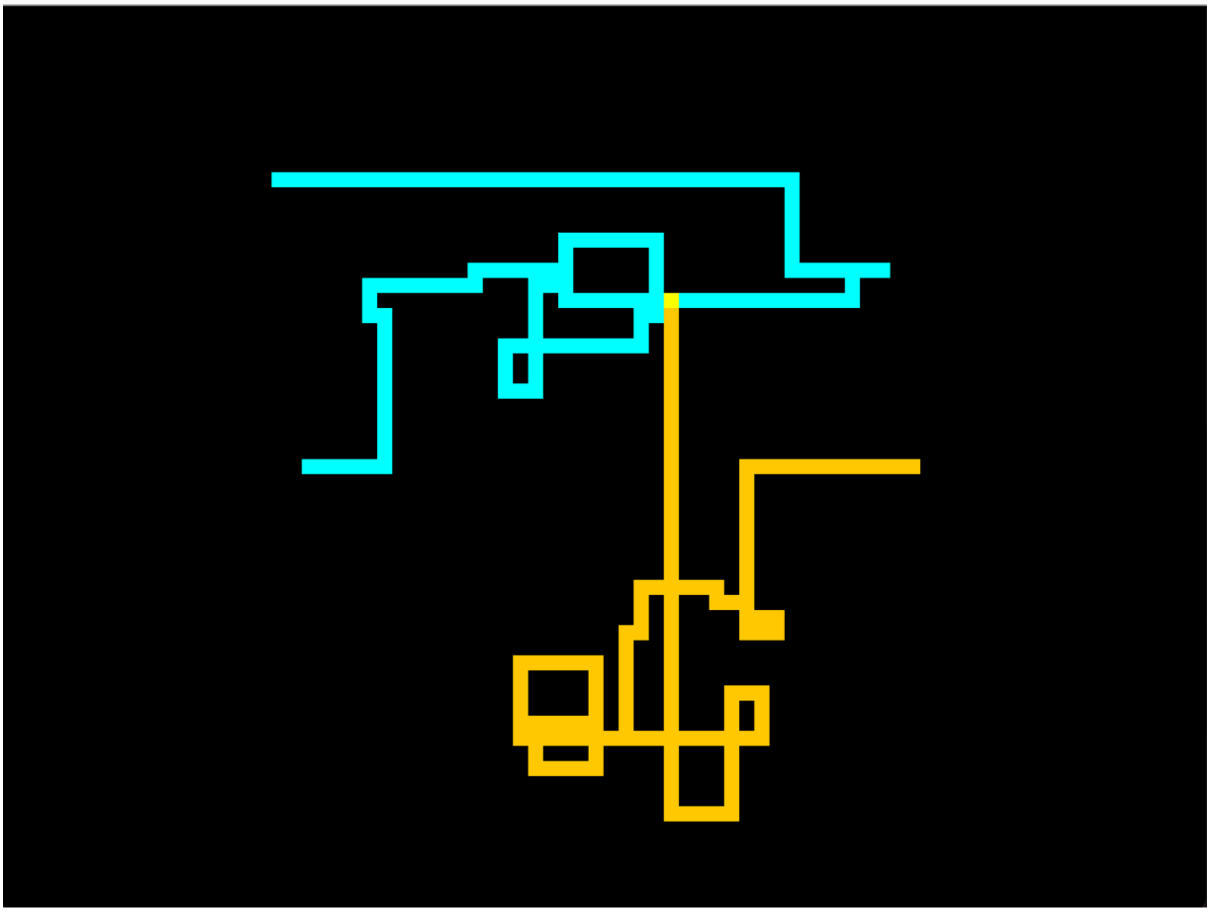
1. Event:moving the game forward
gets handled by the onTick() logic checking it every 80 ms so the game progresses

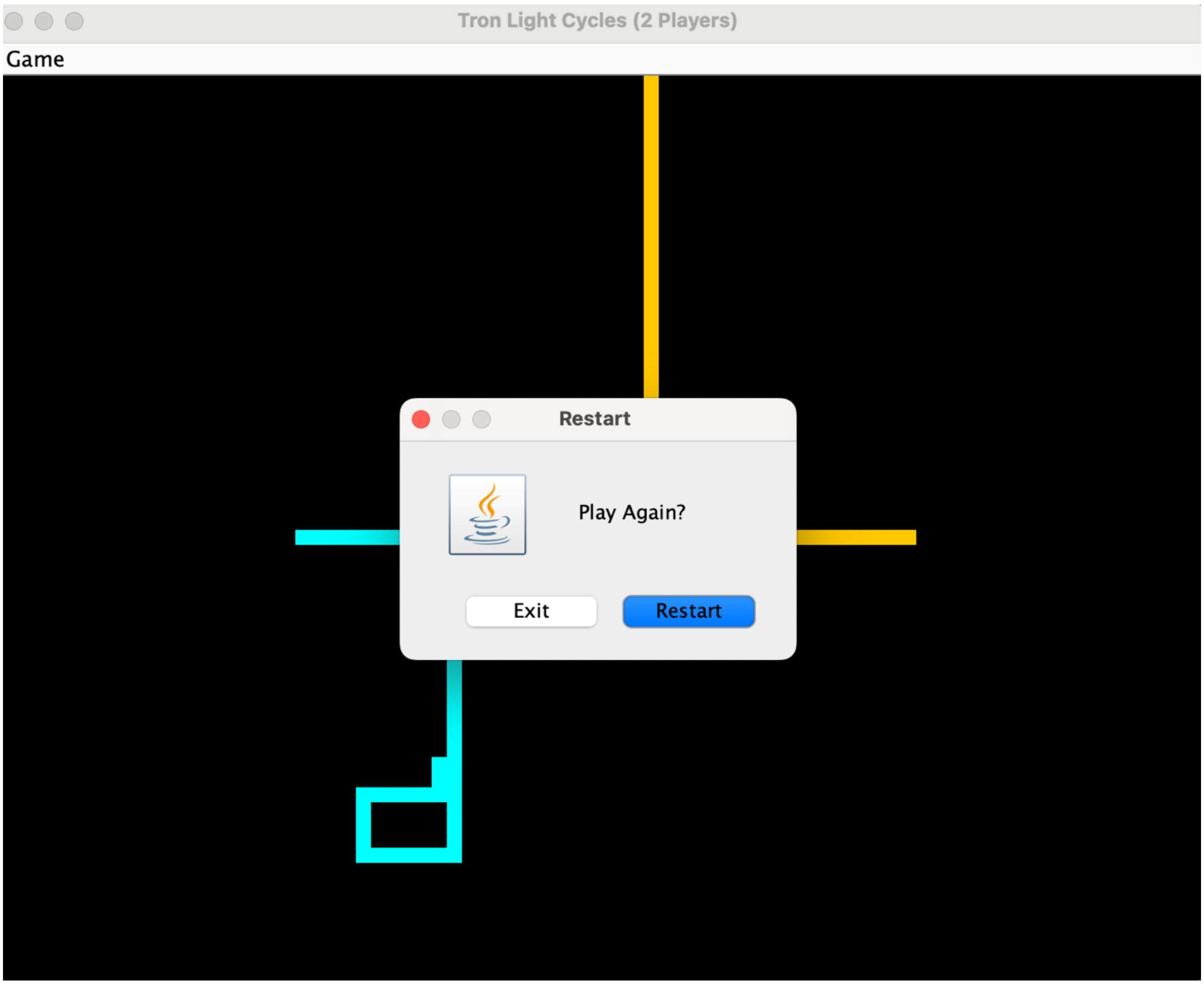
2.pressing buttons to move mottors
handler is setupKeys(): Handles by maping players arrow and wasd keys.
3.event opening menu buttons
Handlers are restartGame(), gui.showHighScores(), System.exit(0);

Test cases:

Features	**Test**	**Result**
Motor movement	**WASD and arrows**	**The motors move**
Bound checking	**Hitting bounds of the board**	**losing**

Trace	Trace hitting	losing
		




Tron Light Cycles (2 Players)
Game
Restart
Play Again?
Exit
Restart